

8085 Assembly Language Programs

Extracted from the uploaded Computer Organization and Microprocessor Architecture Lab Manual

All experiment titles and programs below are transcribed from the manual. No theory, procedure, observation, or result sections are included.

Experiment 1 - Part A

Addition of Two 8-bit Numbers Using 8085

Assembly Language Program

```
LDA 2050H
MOV B, A
LDA 2051H
ADD B
STA 2052H
HLT
```

Experiment 1 - Part B

Addition of Two 16-bit Numbers Using 8085

Assembly Language Program

```
LHLD 2050H  
XCHG  
LHLD 2052H  
DAD D  
SHLD 2054H  
HLT
```

Experiment 1 - Part C

Subtraction of Two 8-bit Numbers Using 8085

Assembly Language Program

```
LDA 2051H  
MOV B, A  
LDA 2050H  
SUB B  
STA 2052H  
HLT
```

Experiment 2 - Part A

Multiplication of Two 8-bit Numbers Using 8085

Assembly Language Program

```
LDA 2050H
MOV D, A
LDA 2051H
MOV C, A
ORA A
MVI A, 00H
JZ STORE
LOOP: ADD D
DCR C
JNZ LOOP
STORE:
STA 2052H
HLT
```

Experiment 2 - Part B

Division of Two 8-bit Numbers Using 8085

Assembly Language Program

```
LDA 2051H
MOV C, A
LDA 2050H
MVI D, 00H
MOV B, A
MOV A, C
ORA A
MOV A, B
JZ DONE
LOOP: CMP C
JC DONE
SUB C
INR D
JMP LOOP
DONE:
STA 2053H
MOV A, D
STA 2052H
HLT
```

Experiment 3

Addition of a Block of 8-bit Data Stored in Memory

Assembly Language Program

```
LXI H, 2050H
MOV C, M
MVI A, 00H
LOOP: INX H
ADD M
DCR C
JNZ LOOP
STA 2055H
HLT
```

Experiment 4 - Part A

Find the Minimum of Two 8-bit Numbers Using 8085

Assembly Language Program

```
LDA 2050H
MOV B, A
LDA 2051H
CMP B
JC STORE
MOV A, B
STORE:
STA 2052H
HLT
```

Experiment 4 - Part B

Find the Minimum from a Block of N 8-bit Numbers Using 8085

Assembly Language Program

```
LXI H, 2050H
MOV C, M
INX H
MOV A, M
DCR C
LOOP: INX H
CMP M
JC SKIP
MOV A, M
SKIP: DCR C
JNZ LOOP
STA 2055H
HLT
```

Experiment 5 - Part A

Find the Maximum of Two 8-bit Numbers Using 8085

Assembly Language Program

```
LDA 2050H
MOV B, A
LDA 2051H
CMP B
JNC STORE
MOV A, B
STORE:
STA 2052H
HLT
```

Experiment 5 - Part B

Find the Maximum from a Block of N 8-bit Numbers Using 8085

Assembly Language Program

```
LXI H, 2050H
MOV C, M
INX H
MOV A, M
DCR C
LOOP: INX H
CMP M
JNC SKIP
MOV A, M
SKIP: DCR C
JNZ LOOP
STA 2055H
HLT
```

Experiment 6 - Part A

Sort Data in Ascending Order Using 8085

Assembly Language Program

```
LDA 2050H
CPI 02H
JC HALT
DCR A
MOV D, A
OUTER:
LDA 2050H
DCR A
MOV C, A
LXI H, 2051H
INNER:
MOV A, M
INX H
CMP M
JC SKIP
JZ SKIP
MOV B, M
MOV M, A
DCX H
MOV M, B
INX H
SKIP:
DCR C
JNZ INNER
DCR D
JNZ OUTER
HALT:
HLT
```

Experiment 6 - Part B

Sort Data in Descending Order Using 8085

Assembly Language Program

```
LDA 2050H
CPI 02H
JC HALT
DCR A
MOV D, A
OUTER:
LDA 2050H
DCR A
MOV C, A
LXI H, 2051H
INNER:
MOV A, M
INX H
CMP M
JNC SKIP
MOV B, M
MOV M, A
DCX H
MOV M, B
INX H
SKIP:
DCR C
JNZ INNER
DCR D
JNZ OUTER
HALT:
HLT
```

Experiment 7 - Part A

Convert BCD Number to Binary Using 8085

Assembly Language Program

```
LDA 2050H
MOV B, A
ANI 0F0H
RRC
RRC
RRC
RRC
MOV C, A
ADD A
ADD A
ADD C
ADD A
MOV C, A
MOV A, B
ANI 0FH
ADD C
STA 2051H
HLT
```

Experiment 7 - Part B

Convert Binary Number to BCD Using 8085

Assembly Language Program

```
LDA 2050H
MVI B, 00H
MVI C, 00H
HUNDREDS:
CPI 64H
JC TENS
SUI 64H
INR B
JMP HUNDREDS
TENS:
CPI 0AH
JC UNITS
SUI 0AH
INR C
JMP TENS
UNITS:
MOV D, A
MOV A, C
RLC
RLC
RLC
RLC
ADD D
STA 2052H
MOV A, B
STA 2051H
HLT
```

Experiment 8 - Part A

Convert Binary Number to ASCII Using 8085

Assembly Language Program

```
LDA 2050H
MOV B, A
ANI 0F0H
RRC
RRC
RRC
RRC
CPI 0AH
JC NUM1
ADI 07H
NUM1:
ADI 30H
STA 2051H
MOV A, B
ANI 0FH
CPI 0AH
JC NUM2
ADI 07H
NUM2:
ADI 30H
STA 2052H
HLT
```

Experiment 8 - Part B

Convert ASCII Number to Binary Using 8085

Assembly Language Program

```
LDA 2050H
SUI 30H
CPI 0AH
JC OK1
SUI 07H
OK1:
RLC
RLC
RLC
RLC
MOV B, A
LDA 2051H
SUI 30H
CPI 0AH
JC OK2
SUI 07H
OK2:
ADD B
STA 2052H
HLT
```

Experiment 9

Calculate the Sum of a Series of Even Numbers Using 8085

Assembly Language Program

```
LXI H, 2050H
MOV C, M
MVI B, 00H
LOOP:
INX H
MOV A, M
ANI 01H
JNZ SKIP
MOV A, B
ADD M
MOV B, A
SKIP:
DCR C
JNZ LOOP
MOV A, B
STA 2055H
HLT
```

Experiment 10

Calculate the Sum of a Series of Odd Numbers Using 8085

Assembly Language Program

```
LXI H, 2050H
MOV C, M
MVI B, 00H
LOOP:
INX H
MOV A, M
ANI 01H
JZ SKIP
MOV A, B
ADD M
MOV B, A
SKIP:
DCR C
JNZ LOOP
MOV A, B
STA 2055H
HLT
```